

Team6 — CS 200 Project 1

Chocoholics Anonymous (ChocAn)

Fall 2026 — Due Wednesday, September 23, 2026, 11:59 p.m.

Submitter: *Aaron Suarez*

GitHub repository: <https://github.com/ahsuarez-ua/CS200-ChocAn-Team-6>

Slack Channel: #tuscaloosans-team-6

1. Header Table

Name	CWID	Crimson Email	Hours Contributed
Jacob Wilson	12343358	jrwilson29@crimson.ua.edu	3 hrs
Aaron Suarez	11922638	ahsuarez@crimson.ua.edu	3 hrs
Gabriel Mckinney	12547356	grmckinney@crimson.ua.edu	2 hrs
Peyton Hardy	12393230	pchardy@crimson.ua.edu	2 hrs

2. Task Distribution Table

Name	Tasks Performed	% Contributed
Aaron Suarez	ChocAn Overview, Glossary, actor identification, submitter	25%
Gabriel Mckinney	Use case diagram in UMLetino	25%
Peyton Hardy	Use case descriptions (provider side)	25%
Jacob Wilson	Use case descriptions (manager/operator side), report assembly	25%
Total		100%

3. Detailed Written Solution

ChocAn is an organization that helps people with chocolate addiction. Members pay a monthly fee that entitles them to unlimited consultations and treatments with health care professionals, mainly dietitians, internists, and exercise experts. Each provider serves ChocAn members through a specially designed ChocAn computer terminal. This team consists of Jacob Wilson, Aaron Suarez, Peyton Hardy, and Gabriel McKinney. Our team is building only the ChocAn data-processing software, using C++ and Java/JavaScript. Another organization provides the communications software, the provider terminal, the software used by Acme Accounting Services, and the EFT implementation. Our software validates members, records and bills services provided to members, produces the Provider Directory, maintains member and provider records, and generates the weekly member, provider, and summary reports along with the EFT data. The identified actors are member, operator, manager, scheduler, and provider.

We made the following assumptions. The provider terminal input is simulated by keyboard, and terminal output appears on screen, as required for the acceptance test. No report is emailed; each report is written to its own file named with the member or provider name followed by the report date. The clock, rather than a person, triggers the weekly accounting run, so we model it with a scheduler actor. The EFT step writes a file containing each provider's name, provider number, and transfer amount. We did not model Acme Accounting Services as an actor because its software is outside the scope of our contract. We modeled record management as a single use case covering both Member and Provider records, and the weekly outputs as a single Generate Reports use case, which a manager can also run on demand. Aaron Suarez will submit this report.

4. Glossary

Term	Definition
ChocAn	Chocoholics Anonymous — healthcare organization that the data-processing software is being built for.
Member	A person who pays a monthly fee and is entitled to consultations and treatments.
Provider	A health care professional who delivers services to members, bills ChocAn using a terminal
Manager	A ChocAn employee who may request any report on demand during the week.
Operator	A ChocAn Data Center employee who adds, deletes, and updates member and provider records interactively. These employees do not handle fee-related suspensions or reinstatements.
Acme Accounting Services	Third-party organization, external to the software being built. Records fee payments and handles suspending and reinstating members. Updates ChocAn records nightly at 9 p.m.
Scheduler	The non-human trigger that starts the main accounting procedure at midnight on Friday.

Term	Definition
Member card	Plastic card embossed with the member's name and nine-digit number, with the same data on a magnetic strip.
Member number	Nine-digit identifier for a member.
Provider number	Nine-digit identifier for a provider.
Service code	Six-digit code identifying a service
Provider terminal	Card-reader device at the provider's site that communicates with the Data Center.
ChocAn Data Center	Central system that verifies member numbers, stores service records, and runs the accounting procedure.
Validated	One-line display message shown when a member number is valid.
Invalid number	Display message shown when the member number does not exist.
Member suspended	Display message indicating fees are overdue by at least a month; member status is suspended until member pays the required fees.
Provider Directory	Alphabetical list of service names with their service codes and fees, produced as a file.
Member report	Weekly per-member list of services received, sorted by service date, written to its own file.
Provider report	Weekly per-provider list of services billed, in the order data were received, with consultation count and weekly fee total.
Summary report	Accounts-payable report listing each provider to be paid, consultation counts, fee totals, and overall totals.
Weekly Accounting Procedure	Process that runs at 12:00 AM every Friday that generates reports based on the week's file of services. The process is triggered by the Scheduler actor.
EFT	Means electronic funds transfer. A file that contains provider name, provider number, and amount to be transferred.
Service record	A record on the disk for a service that has been billed. All of which includes the current date and time, the date of the service, the provider number, the member number, and the service code.

5. Use Case Diagram

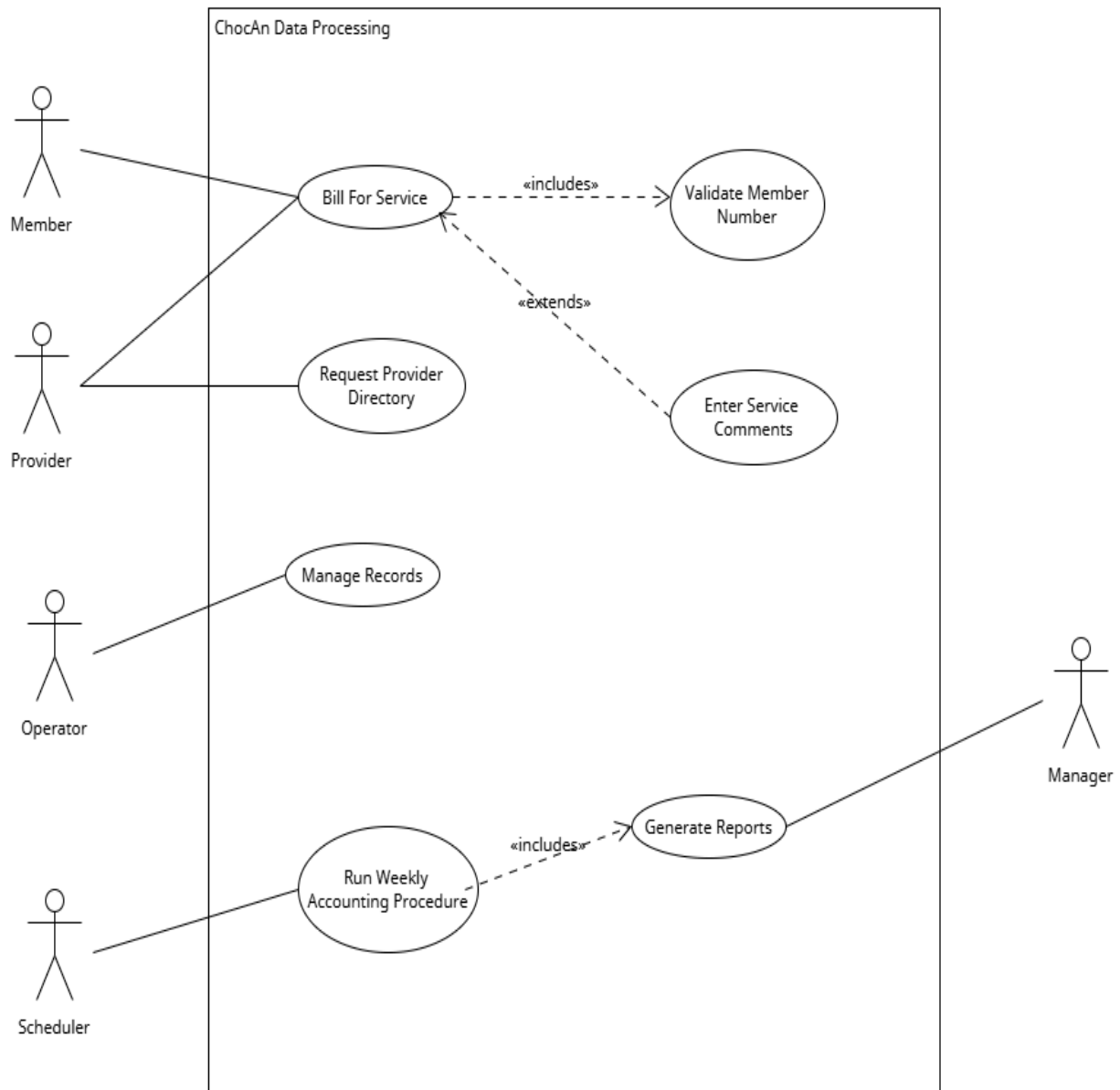


Figure 1. UML use case diagram for the ChocAn data processing software.

5.1 Candidate Actors

- 1) Member - presents their card to a provider to receive services
- 2) Provider - provides services to members and bills ChocAn for them
- 3) Manager - request any report individually at any given time during the week
- 4) Operator - adds/deletes/updates member and provider records
- 5) Scheduler (non-human actor) - triggers the midnight-Friday accounting run

5.2 Candidate Use Cases

- 1) Bill for service
- 2) Validate member number
- 3) Enter service comments
- 4) Request provider directory
- 5) Manage records
- 6) Run weekly accounting procedure
- 7) Generate reports

6. Use Case Descriptions

6.1 Bill for Service

Use Case: Bill for Service

Context: A provider bills ChocAn for a service already rendered to a member.

Actors: Provider

Main Success Scenario:

- 1) The provider passes the member card through the card reader or keys in the member number.
 - a) The system performs *Validate Member Number* and displays Validated.
- 2) Provider keys in the date of service in MM-DD-YYYY format.
- 3) Provider looks up the six-digit service code in the Provider Directory and keys it in.
- 4) The system displays the service name (up to 20 characters) for confirmation.
- 5) The provider confirms the displayed service is correct.
- 6) The system writes a service record to disk containing current date and time, date of service, provider number, member number, service code, and optional comments.
- 7) The system looks up the fee and displays it on the provider's terminal.
- 8) Use case ends; provider records the transaction on the verification form.

Extensions:

- 1) 1a. The member number is invalid. The system displays Invalid number and the use case terminates.
- 2) 1b. The member is suspended. The system displays Member suspended and the use case terminates.
- 3) 3a. The service code does not exist. The system prints an error message and returns to step 3.
- 4) 5a. The provider indicates the displayed service is wrong. The system returns to step 3.
- 5) 5b. The provider chooses to add comments. *Enter Service Comments* executes, then flow resumes at step 6.

6.2 Request Provider Directory

Use Case: Request Provider Directory

Context: A provider asks the system for the Provider Directory and an alphabetically ordered list of service names with their corresponding service codes and fees. This allows them to look up the correct six-digit code for a service.

Actors: Provider

Main Success Scenario:

- 1) The provider requests the Provider Directory from the terminal.
- 2) The system retrieves the directory and displays the services in alphabetical order, with each service's code and fee.
- 3) The provider finds the needed service code and closes the directory. Then the use case ends.

Extensions:

- 1) 2a. The directory cannot be retrieved. The system displays an error message, and the use case ends.
- 2) 3a. The provider cannot find the needed service. The provider closes the directory without selecting a code.

6.3 Manage Records

Use Case: Manage Records

Context: An operator at the ChocAn Data Center works in interactive mode during the day to add, delete, or update the member and provider records.

Actors: Operator

Main Success Scenario:

- 1) The operator selects whether to work with a member record or a provider record.
- 2) The operator selects the operation to perform: add, delete, or update.
 - a) To add, the operator enters the new record's details and the system stores the record.
 - b) To delete, the operator enters the number of the record and the system removes it.
 - c) To update, the operator enters the number of the record, the system displays the current details, and the operator enters the new values.
- 3) The system writes the change to the ChocAn Data Center records.
- 4) The system confirms the change to the operator. (End)

Extensions:

- 1) **2a)** The record number entered for a delete does not match an previous record. The system displays an error and returns to step 2.
- 2) **2b)** The record number entered for an update does not match a previous record. The system displays an error and returns to step 2.
- 3) **2c)** The operator cancels before the change is written. The change is not saved and the use case terminates.
- 4) **3a)** The record cannot be written, so the system displays an error. The change is not saved and the use case terminates.

6.4 Run Weekly Accounting Procedure

Use Case: Run Weekly Accounting Procedure

Context: The main accounting procedure runs at midnight on Friday at the ChocAn Data Center. It reads the week's services file and produces the weekly reports and electronic funds transfer data.

Actors: Scheduler

Main Success Scenario:

- 1) The scheduler triggers the accounting procedure at midnight on Friday.
- 2) The system reads the week's services file provided.
- 3) The system runs Generate Reports, producing the member reports, the provider reports, and the accounts payable summary report.
- 4) The system writes a record of electronic funds transfer data to disk, containing each provider's name, provider number, and the amount to be transferred.
- 5) The system confirms completion of the accounting run. (End)

Extensions:

- 1) **2a)** The week's service file contains no records. The system does not create any member or provider reports. It generates a summary report showing zero providers, zero consultations, and a fee total of zero, then continues at step 4.
- 2) **2b)** The week's service file cannot be read. The system reports the failure, does not produce any output, and the use case ends.
- 3) **4a)** The EFT file cannot be written. The system reports the failure, and the use case ends without generating any EFT data.

6.5 Generate Reports

Use Case: Generate Reports

Context: The system produces the weekly reports: one for each member who consulted a provider, one for each provider who billed ChocAn, and a summary report for accounts payable. Runs as part of the weekly accounting procedure. Can also be requested by a manager at any time during the week.

Actors: Manager

Main Success Scenario:

- 1) The system reads the week's service file.
- 2) Each week, if a member visits a provider, the system creates a report for them. This report includes the member's name, number, and address, along with the city, state, and ZIP code. For every service they received, it lists the date, provider name, and service name, all sorted by the date of service.
- 3) The system saves each member's report as its own file, using the member's name and the report date in the filename.
- 4) Each week, if a provider bills ChocAn, the system creates a report just for them. This report includes the provider's name, number, and address, as well as the city, state, and ZIP code. For every service they've given, the report lists when the service happened, when the system received the data, the member's name and number, the service code, and the fee. Everything is listed in the order the information came in.
- 5) At the end of each week, the system adds up the total number of consultations and the total fees for each provider. It includes these totals in the provider's report and then saves the report as a file. The file is named using the provider's name and the date of the report.
- 6) At the end of the week, the system puts together a summary report. This report lists every provider who's getting paid, how many consultations each one had, and each provider's total fee. After that, it shows the total number of providers who gave services, the total number of consultations, and the overall fee total for the week.
- 7) The system confirms the reports have been produced. (End)

Extensions:

- 1) **1a)** The week's service file cannot be read. The system reports the failure, produces no reports, and the use case terminates.
- 2) **2a)** No member consulted a provider during the week. The system produces no member reports and continues at step 4.
- 3) **4a)** No provider billed ChocAn during the week. The system produces no provider reports and continues at step 6.
- 4) **3a)** A report file cannot be written. The system reports the failure for that report and continues with the remaining reports.

6.6 Validate Member Number

Use Case: Validate Member Number

Context: The system checks a member number entered at a provider terminal before the provider bills for a service.

Actors: Provider

Main Success Scenario:

- 1) The provider scans the member card or enters the nine-digit member number.
- 2) The system finds the member record and checks the member's status.
- 3) The system displays "Validated." The provider may continue with Bill for Service.

Extensions:

- 1) 2a) The member number does not exist. The system displays "Invalid number," and the use case ends.
- 2) 2b) The member is suspended. The system displays "Member suspended," and the use case ends.

6.7 Enter Service Comments

Use Case: Enter Service Comments

Context: A provider optionally adds comments to a service record while billing ChocAn.

Actors: Provider

Main Success Scenario:

- 1) After confirming the service name, the provider chooses to add comments.
- 2) The provider enters the comments.
- 3) The system includes the comments in the service record and returns to Bill for Service.

Extensions:

- 1) 2a) The provider cancels the comment entry. The system discards the unfinished comments and returns to Bill for Service without comments.